

THE MUSCULOSKELETAL SYSTEM

Musculoskeletal refers to the muscles and the skeleton. The musculoskeletal system **protects, supports**, aids in the **movement** of body parts, and essential in *hematopoiesis* or **blood production** (*hemat/o* = blood, *-poiesis* = production). The bones also **store fat** in the bone marrow and to store and release **minerals**.

CARTILAGE

Cartilage is a dense, elastic connective tissue that covers and protects the ends of long bones. The medical term ***chondral*** refers to cartilage

MUSCLES & ASSOCIATED STRUCTURES

Muscles consist of cells and fibers that contract and produce body movement. The ***three types*** of muscles are:

I-Cardiac muscle (heart muscle)

II-Smooth muscle known as visceral or involuntary muscle; found in the internal organs such as the stomach

III-Skeletal muscle the only muscle that can be controlled voluntarily & attached to the bones.

Fascia is a fibrous tissue that's responsible for covering, supporting, separating, and allowing muscles to slide easily.

Tendons, are tissue fibers that connect the muscles to the bone.

Articulation, commonly known as ***joint***, is the connection made between bones. Articulations **have cavities** between adjoining bones are known as ***synovial joints***.

Ligaments are bands of connective tissues responsible for connecting bones and cartilages and, at the same time, are essential in supporting and strengthening joints.

DISEASES, DISORDERS, & DIAGNOSTIC TERMS in MUSCULOSKELETAL SYSTEM

Causes of musculoskeletal problems are injury, infections, malignancies, and connective tissue disorders.

Fractures and dislocations are the most common traumatic injuries sustained by the bones. Cord injury may cause *paralysis* or *paresis* below the injured part.

Paraplegia: paralysis of both legs

Quadriplegia/tetraplegia: paralysis of upper and lower extremities

Paraparesis: weakness of the both legs

Quadriparesis/tetraparesis: weakness of all four extremities

Remember the suffix “*itis*” means inflammation.

List of infections affecting the musculoskeletal system:

cellul/o (little cell) + *itis* = *cellulitis*

my/o (muscle) + *itis* = *myocellulitis*

oste/o (bone) + *itis* = *osteitis*

Alterations with metabolism can lead to bone destruction and problems with bone formation. *Example:*

malacia (softening) = *osteomalacia*

-porosis (thinning) = *osteoporosis*

Any disease of the joint is known as *arthropathy*.

Remember that *arthr/o* refers to the joint.

Arthroscopy

Arthroscope

Arthrotomy

Arthritis

SURGICAL AND THERAPEUTIC INTERVENTIONS

Orthopedic surgeons perform surgical procedures to restore bones to their normal function.

Common surgical procedures associated with the musculoskeletal system:

vertebroplasty	vertebro + plasty	repair of the vertebrae
myoplasty	myo + plasty	repair of muscle
chondrectomy	chondro + ectomy	excision of a cartilage
arthrocentesis	arthro + centesis	extraction of accumulated fluid in the synovial joint

Note that *anti-* is a prefix meaning against, thus:

Anti-inflammatories: drugs that reduce inflammation and pain.

Anti-arthritis: drugs that provide relief from arthritic symptoms.

The circulatory system

Is made up of the *cardiovascular system* and the *lymphatic system*. This body system functions to maintain *homeostasis*.

CARDIOVASCULAR SYSTEM (CVS),

The *cardiovascular* (*cardio* + *vascul* + *ar*) system consists of the heart and the blood vessels.

Heart

The heart has four chambers:

Right and left *atria* (plural for *atrium*)

Right and left *ventricles*

<i>tricuspid</i>	tri (three) + cuspid	valve on the right side of the heart
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<i>bicuspid</i>	bi (two) + cuspid	valve on the left side of the heart, otherwise known as <i>mitral</i> valve
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The heart wall is made up of several layers:

Epicardium: outer layer

Myocardium: muscular middle layer

Endocardium: inner layer

Blood Vessels

Arteries: carry oxygenated blood from the heart to the different body parts

Arterioles: small blood vessels that connect the arteries to the capillaries

Capillaries: smallest blood vessels where oxygenation and nutrients occur

Venules: small blood vessels that connect the capillaries to the veins

Veins: carry deoxygenated blood from the different parts of the body to the heart

Note that the *pulmonary artery* is the only artery that carries deoxygenated blood and the

pulmonary vein is the only vein that carries oxygenated blood.

The *coronary arteries* carry oxygenated and nutrient-filled blood to the *myocardium* (heart muscle).

Diseases, Disorders, and Diagnostic Terms in CVS

Cardiomyopathy is the disease of the heart muscle. Example:

Myocarditis: inflammation of the myocardium (heart muscle)

Most common diagnostic tests include:

Electrocardiogram electr/o + cardi/o + -gram (*ECG*)

Electrocardiography: a noninvasive procedure used to record electrical activity of the heart

Echocardiogram echo + cardi/o + -gram record of the heart's sonogram

Cardiac catheterization is a procedure in which a *catheter* is moved through a blood vessel to the heart to diagnose heart disorders

Vasoconstriction: decrease in the diameter of a blood vessel

Vasodilation: increase in the diameter of a blood vessel

Surgical & therapeutic intervention

Cardiopulmonary bypass: a technique that temporarily takes over the function of the heart and the lungs during surgery

Cardiopulmonary resuscitation: a series of lifesaving actions that improve a client's chances of survival after a cardiac arrest

LYMPHATIC SYSTEM

STRUCTURES OF THE LYMPHATIC SYSTEM

Lymph: fluid that circulates in the lymphatic system

Lymph vessels: thin tubes that carry lymph and white blood cells

Lymph nodes: small round glands that filter lymph throughout the body

Word parts related to the lymphatic system:

Word Part	Meaning	Word Association
adenoid/o	adenoids	<i>Adenoidectomy</i> is the surgical removal of the adenoids.
lymphat/o	lymphatics	A disease or enlargement of the lymph nodes is <i>lymphadenopathy</i> .
splen/o	spleen	<i>Splenomegaly</i> is the enlargement of the spleen.
lymphoma	lymph + oma	<i>cancer that originates from the lymphatic system</i>
lymphadenoma	lymph + aden + oma	<i>tumor of the lymph node</i>
lymphadenitis	lymph + aden + itis	<i>inflammation of the lymph nodes</i>

Any disease involving the lymph nodes is referred to as ***lymphadenopathy***.

Surgical and Therapeutic Intervention as Lymphadenectomy, Splenectomy, Tonsillectomy, & Adenoidectomy.